

ZEYU HU

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EDUCATION

University of Connecticut	09/2023 – 06/2027 (Expected)
Ph.D. in Statistics	Storrs, CT, US
Dissertation Advisors: Dr. Jun Yan & Dr. Wenrui Li	
Boston University	09/2020 – 05/2022
MA in Statistics	Boston, MA, US
Sun Yat-Sen University	09/2016 – 06/2020
BS in Applied Statistics	Guangzhou, Guangdong, China

RESEARCH

Research Assistant, University of Connecticut Health Center 10/2023 – now

- Conducted spatiotemporal analysis of high-frequency mobility and multiple categories of consumption data during COVID-19 in Seoul, South Korea, deriving elasticity estimates that quantify behavioral responses under varying policy regimes.
- Modeled dynamic interactions among COVID-19 incidence, human mobility, and social distancing policies using district-level VAR models to capture feedback mechanisms across time.
- Constructed complex networks from fine-grained dong-level mobility data to investigate intra- and inter-district connectivity patterns in Seoul during the COVID-19 pandemic.
- Analyzed longitudinal long COVID cohort data using inverse probability of censoring weighting (IPCW) and generalized mixed-effects linear models to assess the impacts of comorbidities on symptom persistence.

PhD Research, Department of Statistics, University of Connecticut 09/2023 – now

- Conducted a comprehensive network analysis of U.S. otolaryngologists using co-authorship data to uncover collaboration patterns and specialty-driven communities.
- Developed a novel clustering methodology that integrates node-level covariates with network topology, and implemented it in a user-friendly R package for broad dissemination in applied research.
- Developed a robust Bayesian noisy-network community detection framework that integrates node-level covariates, and applied it to functional brain networks from the Human Connectome Project and Alzheimer's disease cohorts.
- Implemented Python scripts to harvest large-scale scholarly metadata from Crossref, constructing co-authorship and co-citation networks for further analysis.

INTERNSHIP

Insmmed Incorporated, Biostatistics Intern, Bridgewater, NJ 06/2026 – Present

- Developed interactive Shiny applications to evaluate Bayesian clinical trial designs, including operating characteristics, effective sample size, type I error, power, posterior decision probabilities, and other simulation-based performance metrics under different design scenarios.

- Implemented and compared Bayesian borrowing methods for incorporating historical control information, including approximate normal-based approaches, exact algorithm-based approaches, and dual-model Shiny applications for validating operating characteristic and effective sample size results across alternative model specifications.
- Conducted time-to-event simulation studies to evaluate Bayesian survival analysis methods under different assumptions and summarize design performance.
- Performed data analysis and result interpretation to assess the consistency, robustness, and practical implications of Bayesian design outputs for internal statistical review.

Chinese Evidence-based Medicine Center and Cochrane China Center, West China Hospital, Sichuan University, Chengdu 05/2021 – 07/2021

- Used SQL to clean and classify raw hospital data and established a centralized database for storage.
- Applied regular expressions to determine whether a patient had sepsis based on medication and clinical symptoms, and stored the results in the database.
- Visualized image processing of complex statistical methods, helping doctors to understand and apply them to clinical cases.

Agricultural Bank of China | Big Data Engineer 06/2019 – 08/2019

- Built the data classification and search platform.
- Applied K-means clustering and spectral clustering algorithm to merge tables, made it easier for staff to find information from massive data and significantly reduced storage requirements.

China Telecom, Taiyuan Branch 01/2017 – 02/2017

- Processed background data for BESTPAY (the third-largest mobile payment platform in China) in Shanxi province.
- Participated in business approval procedures and audited annual reports.
- Promoted the adoption of BESTPAY in business operations.

PRESENTATION

- Gave an invited talk at the COVID-19 Data and Disease Modeling International Symposium (Incheon, South Korea, May 2024) on the impact of social distancing policies on mobility and encounter dynamics in Seoul.
- Chaired the session titled “Emerging Topics and Recent Advances in Methods for Survival Analysis” at the 2025 International Chinese Statistical Association (ICSA) Annual Meeting, featuring talks on noisy labels, conformal survival tests, zero-truncated recurrent events, and high-dimensional causal inference.

SKILLS

- R, Python, SAS, SQL, Excel, $\text{\LaTeX}$

PUBLICATIONS

- Hu, Z., Yu, M., Jang, Y., Shrestha, S., Jung, J., & Jo, Y. (2026). Complex Network Analysis of Mobility Dynamics in Seoul During the COVID-19 Pandemic 2020–2022. *Scientific Reports*, 16(1), 4345.
- Hu, Z., Chang, W., Jang, Y., Jo, Y., & Jung, J. (2025). Dynamic Interactions of COVID-19 Incidences, Mobility, and Social Distancing Policies in Seoul: A VAR Model Approach. *Scientific Reports*, 15(1), 24288.
- Jo, Y., Hu, Z., Joo, H., Jung, J., & Lee, J. Time to Recovery from Long COVID: A Longitudinal Analysis of Symptom Duration and Risk Factors Using Accelerated Failure Time Models. *International Journal of Infectious Diseases* (In press).
- Hu, Z., Li, W., Yan, J., & Zhang, P. Covariate Connectivity Combined Clustering for Weighted Networks. *Statistical Papers* (Revise and Resubmit).
- Hu, Z., Chang, W., Jang, Y., Jo, Y., & Jung, J. The Impact of COVID-19 and Social Distancing Policy on Mobility and Expenditure in Seoul. *BMC Public Health* (Revise and Resubmit).
- Lei, B., Li, Y., Zeng, Y., Ren, T., Luo, Y., Shi, T., Gao, Z., Hu, Z., Kang, W., & Chen, Q. (2024). Infant Agent: A Tool-Integrated, Logic-Driven Agent with Cost-Effective API Usage. *arXiv preprint arXiv:2411.01114*.
- Hu, Z., Zhang, P., Yan, J., & Li, W. Robust Community Detection for Noisy Networks with Covariates. (Ongoing)